

WHAM-XREX-PFMG474 Simulator Validation Report

Generated by `python/run_simulator_validation.py`

September 18, 2026

Purpose

Autonomous end-to-end validation of the controller/Transrex-simulator closed loop, per Part C/D of the simulator plan. Each section below covers one test scenario: what was commanded, the expected behavior, the observed result, and a pass/fail verdict. Raw data for every scenario is archived under `shots/` in this repository (gitignored).

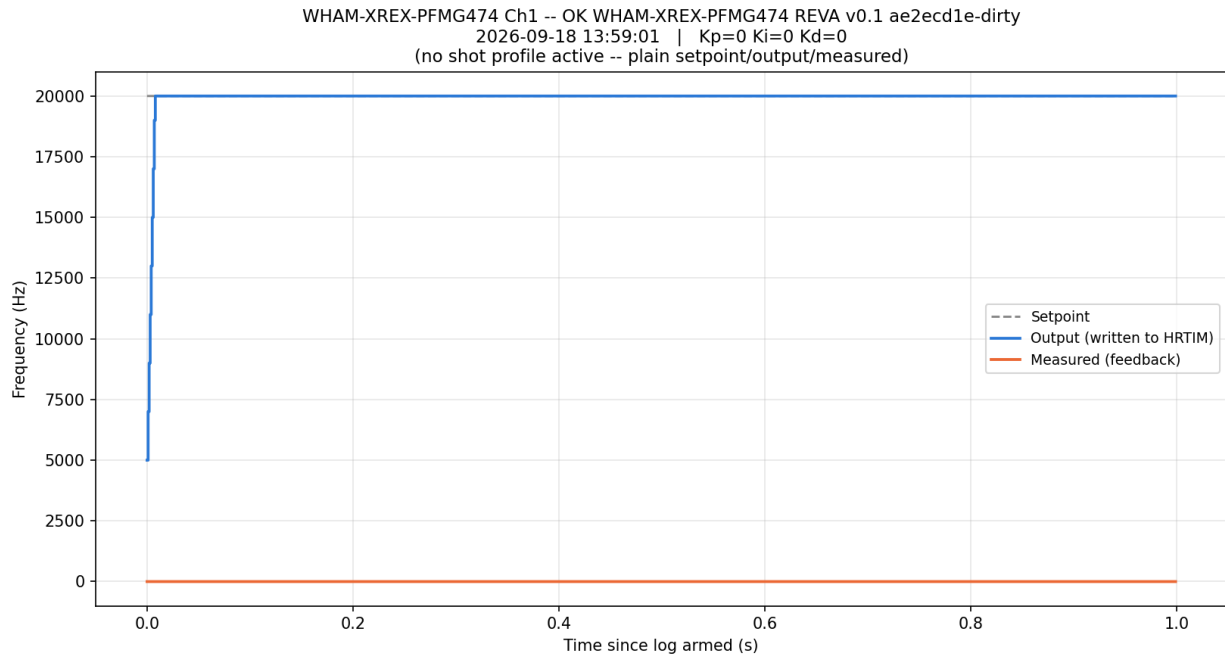
1 Baseline closed-loop tracking

Expected: Simulator DRIVE_HZ converges to within 500 Hz of the commanded 20000 Hz open-loop setpoint

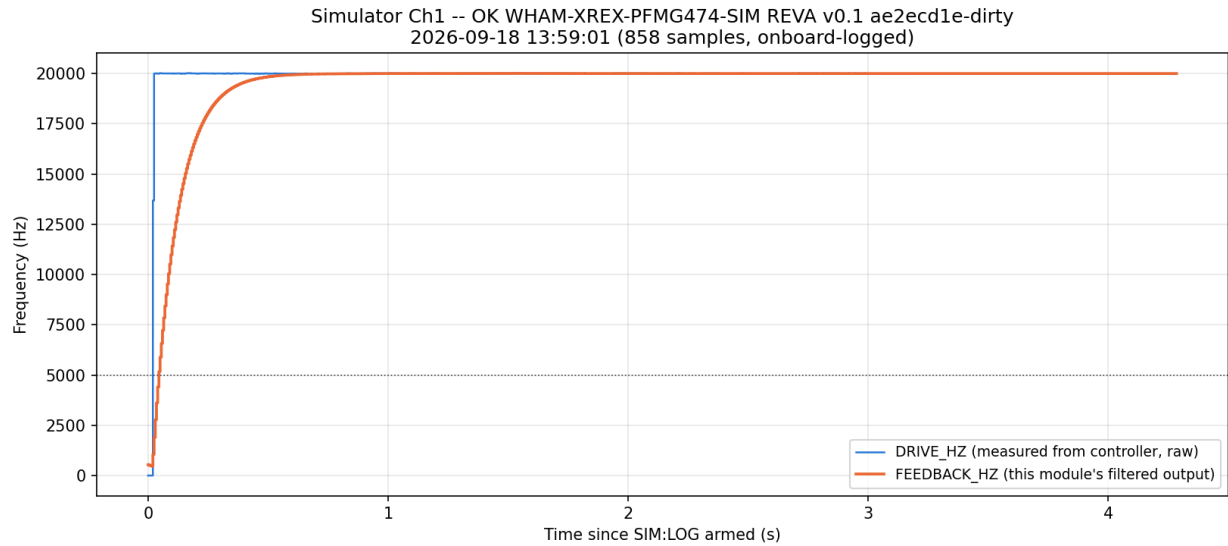
Verdict: PASS

Notes: Final measured DRIVE_HZ=20000 (target 20000, 858 simulator samples / 1000 controller samples over 3.0s).

Raw data: `shots/20260918_135855_baseline/ctrl_ch1.csv`



Raw data: `shots/20260918_135855_baseline/sim_ch1.csv`



2 Full profiled shot

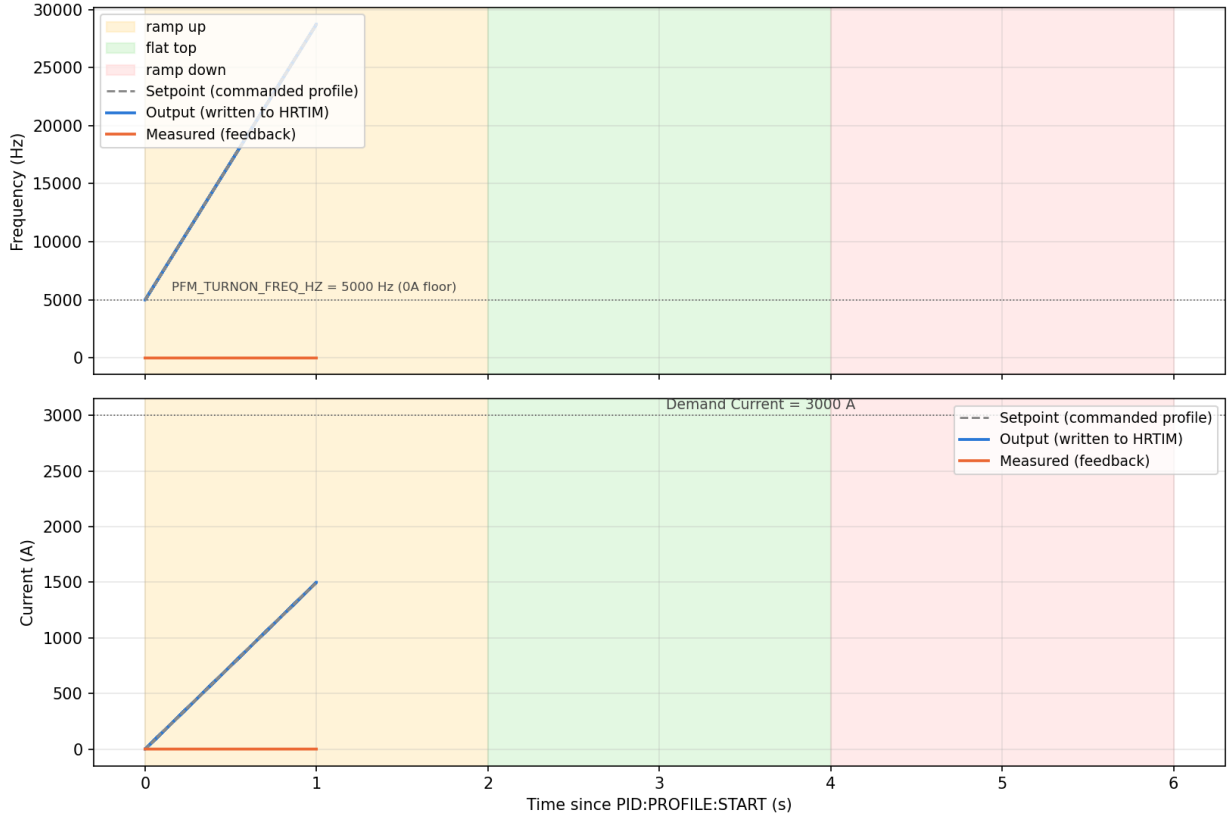
Expected: Controller output tracks the commanded profile (ramp=2.0s flat=2.0s demand=3000.0A) within 2000 Hz max error

Verdict: PASS

Notes: output_error max_abs=0 Hz, 1000 samples @ 1000 Hz.

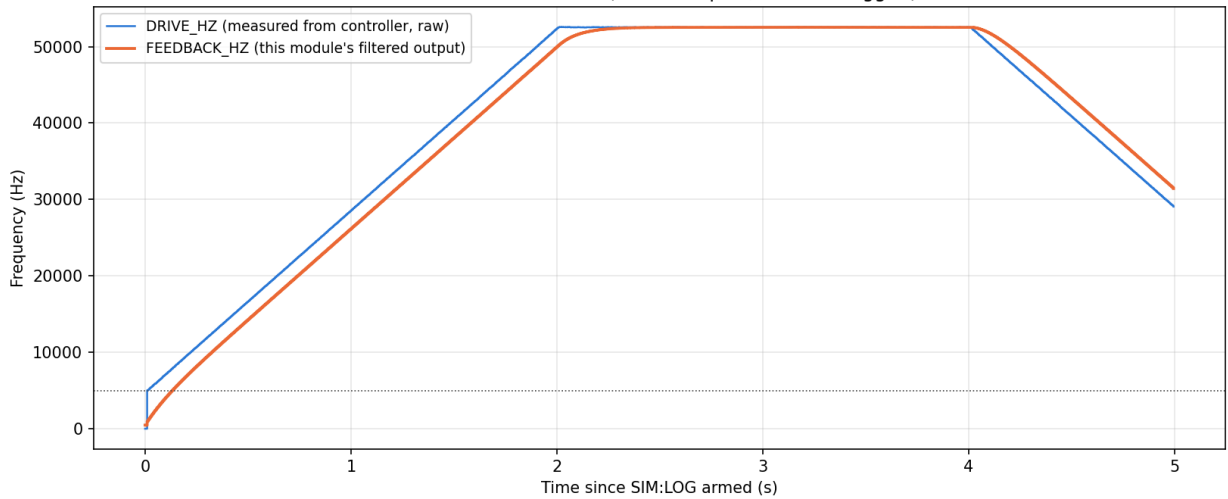
Raw data: shots/20260918_135902_profiled_shot/ctrl_ch1.csv

WHAM-XREX-PFMG474 Ch1 -- OK WHAM-XREX-PFMG474 REVA v0.1 ae2ecd1e-dirty
 2026-09-18 13:59:11 | Kp=0 Ki=0 Kd=0
 Ramp=2.00s FlatTop=2.00s Demand=3000A Total=6.00s



Raw data: shots/20260918_135902_profiled_shot/sim_ch1.csv

Simulator Ch1 -- OK WHAM-XREX-PFMG474-SIM REVA v0.1 ae2ecd1e-dirty
 2026-09-18 13:59:11 (1000 samples, onboard-logged)



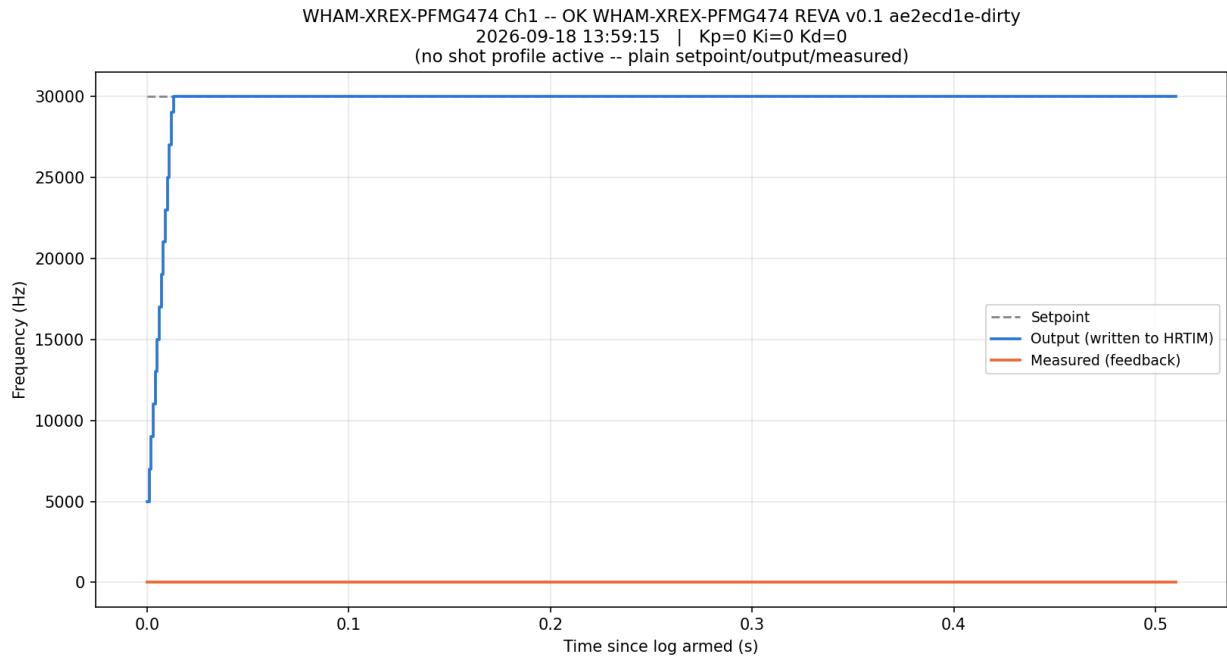
3 OCP fault mid-shot

Expected: STATE? reports OVERCURRENT, survivors keep running at a derated setpoint

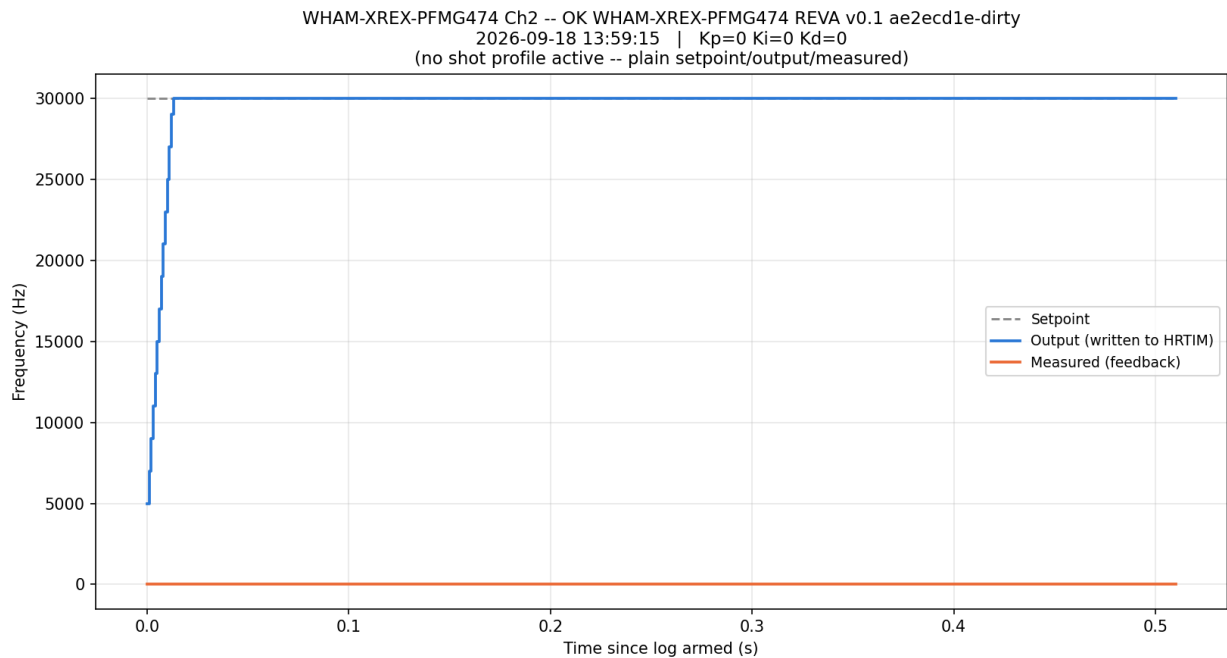
Verdict: PASS

Notes: STATE? -> 'OK FAULT OVERCURRENT 1'; faulted ch1 and survivor ch2 logged (511/511 samples).

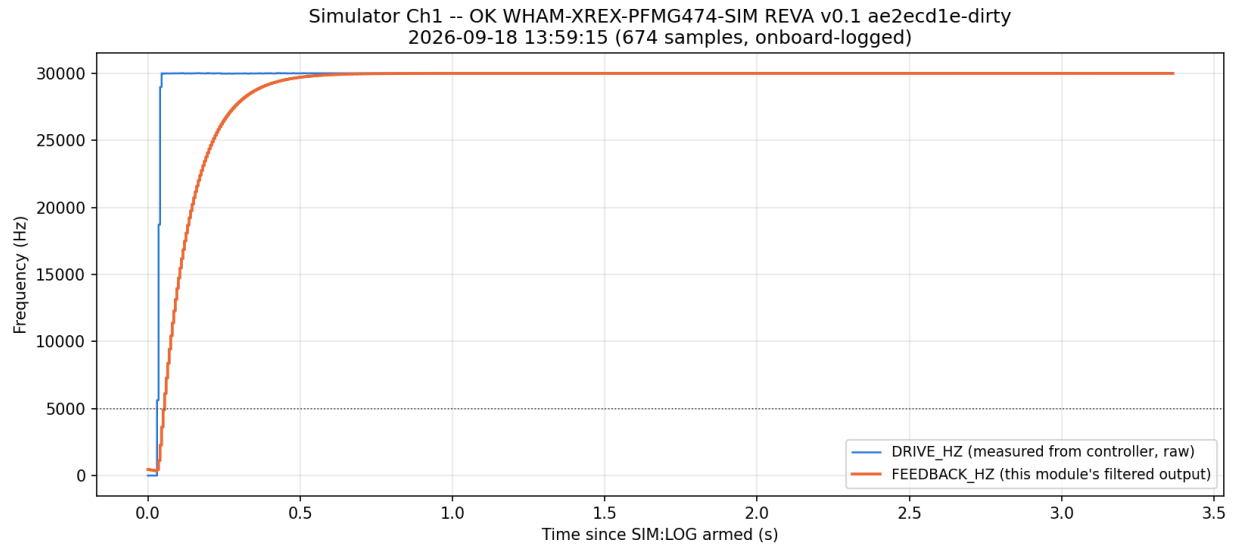
Raw data: shots/20260918_135913_ocp/ctrl_ch1.csv



Raw data: shots/20260918_135913_ocp/ctrl_ch2.csv



Raw data: shots/20260918_135913_ocp/sim_ch1.csv



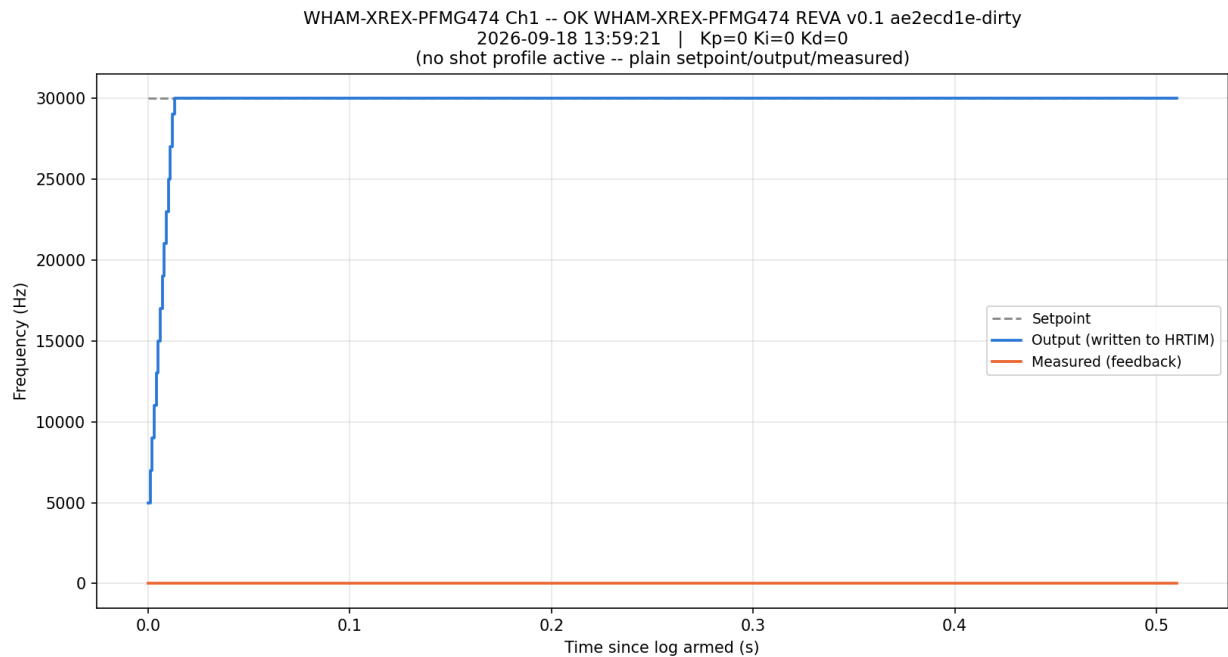
4 Enerpro fault mid-shot

Expected: STATE? reports ENERPRO, survivors keep running at a derated setpoint

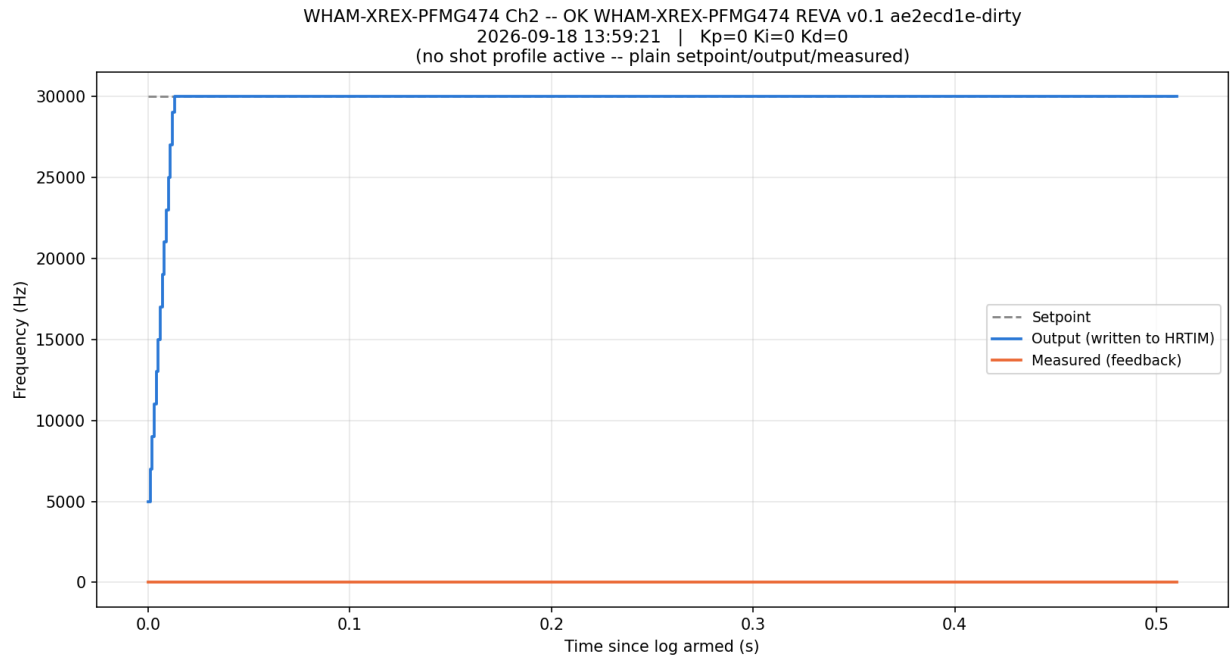
Verdict: PASS

Notes: STATE? -> 'OK FAULT ENERPRO 1'; faulted ch1 and survivor ch2 logged (511/511 samples).

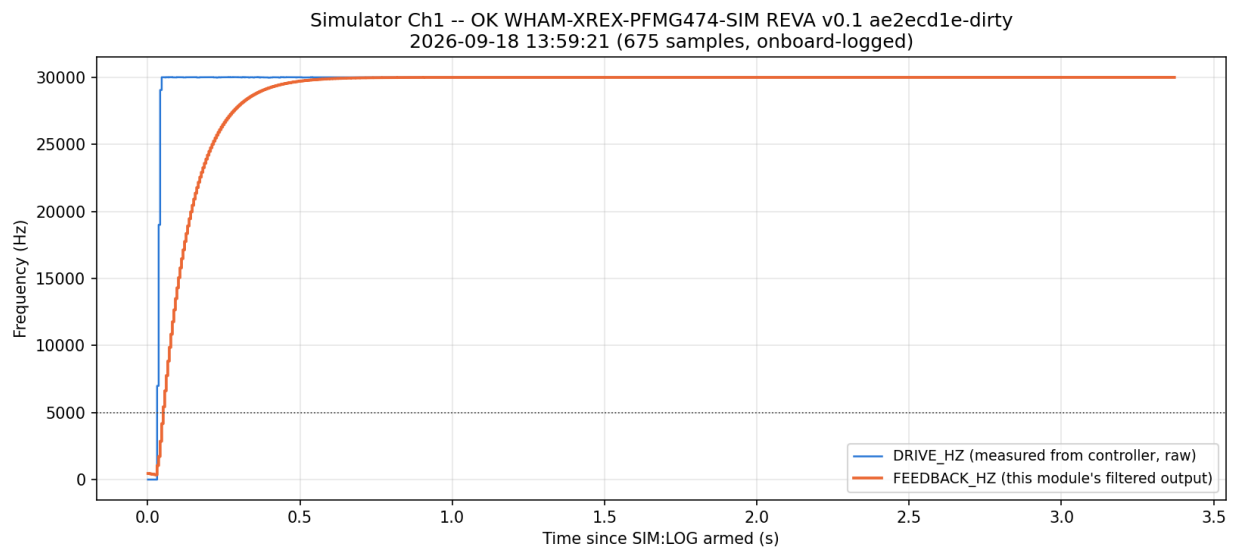
Raw data: shots/20260918_135918_enerpro/ctrl_ch1.csv



Raw data: shots/20260918_135918_enerpro/ctrl_ch2.csv



Raw data: shots/20260918_135918_enerpro/sim_ch1.csv



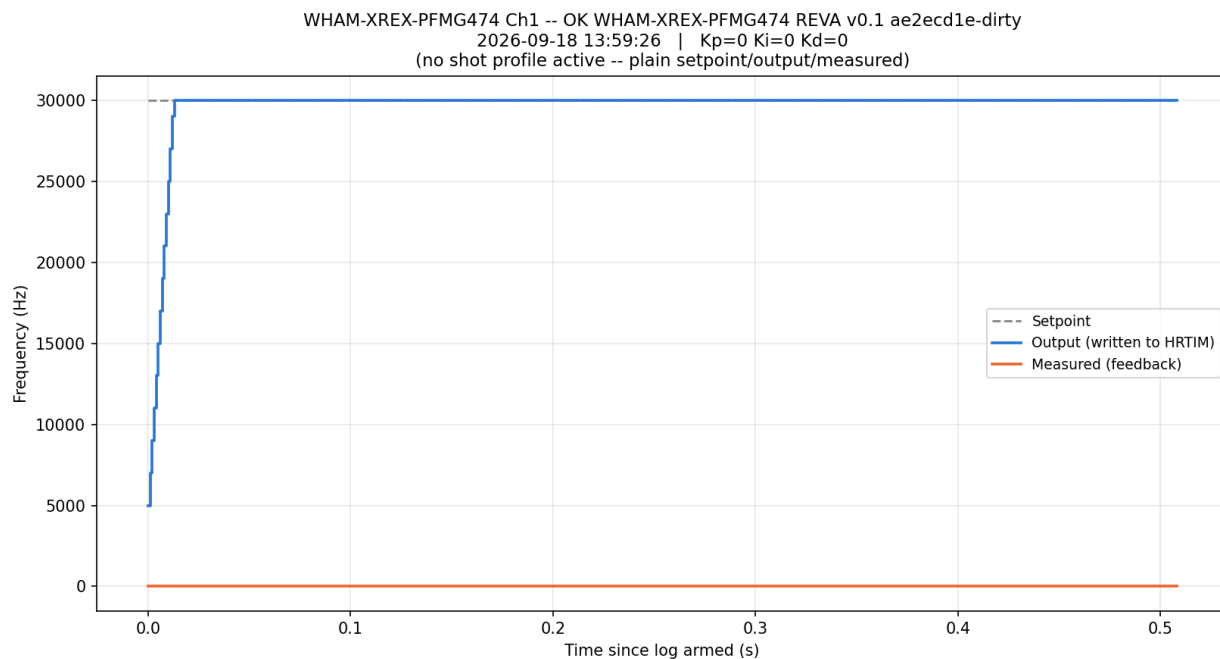
5 Water/Temp fault mid-shot

Expected: STATE? reports GENERAL, full system stop expected

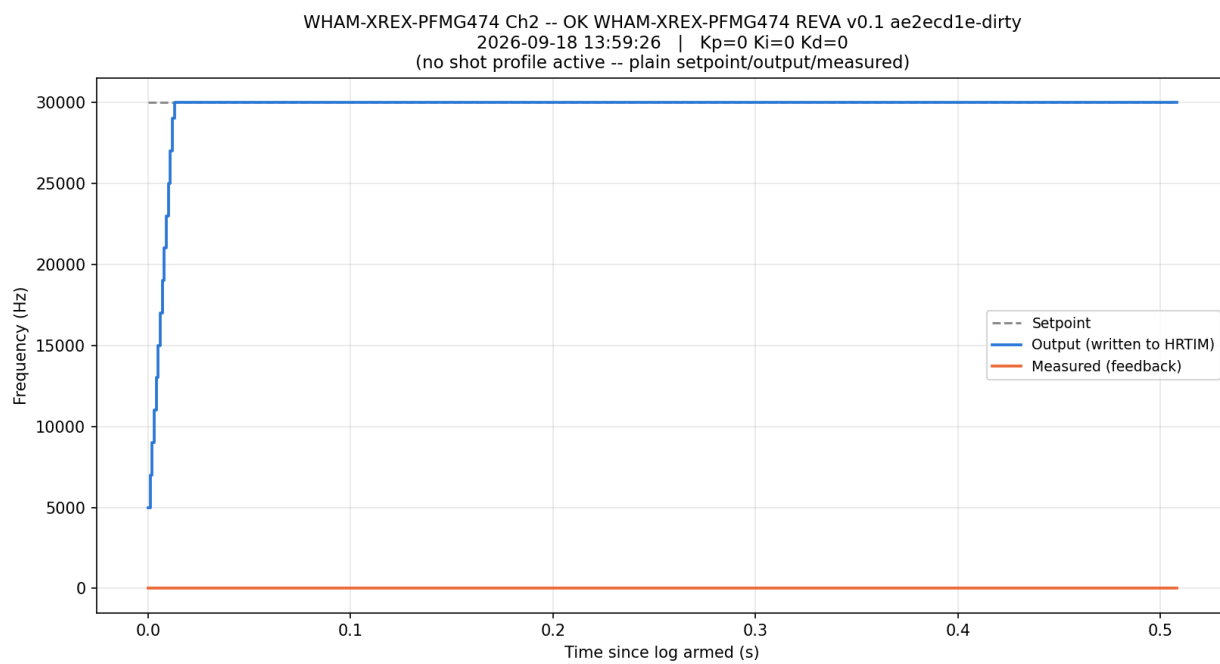
Verdict: PASS

Notes: STATE? -> 'OK FAULT GENERAL'; faulted ch1 and survivor ch2 logged (509/509 samples).

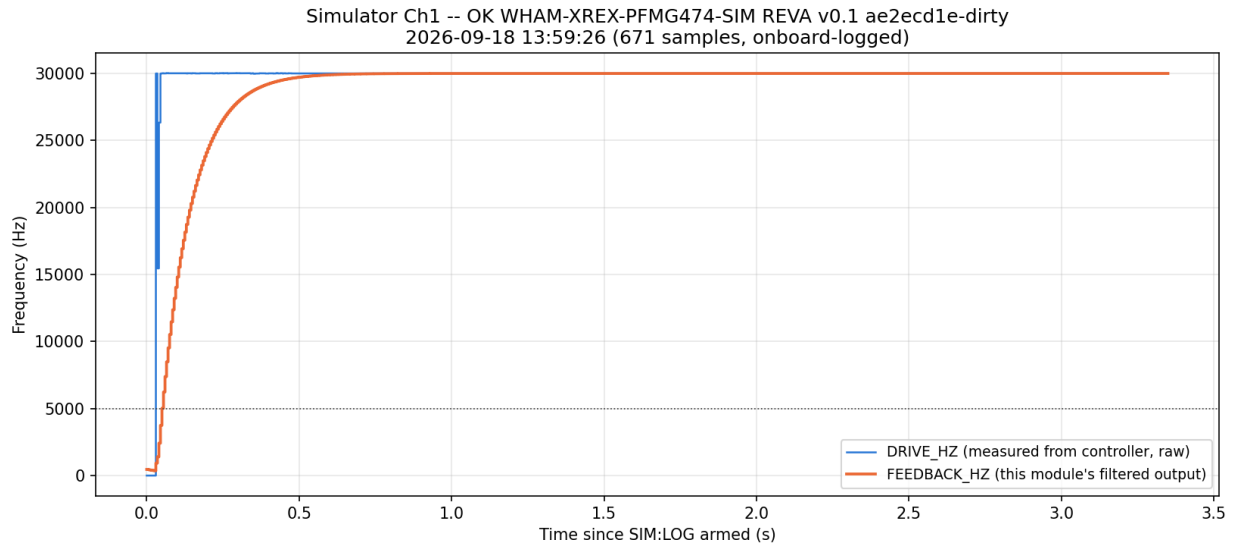
Raw data: shots/20260918_135924_watertemp/ctrl_ch1.csv



Raw data: shots/20260918_135924_watertemp/ctrl_ch2.csv



Raw data: shots/20260918_135924_watertemp/sim_ch1.csv



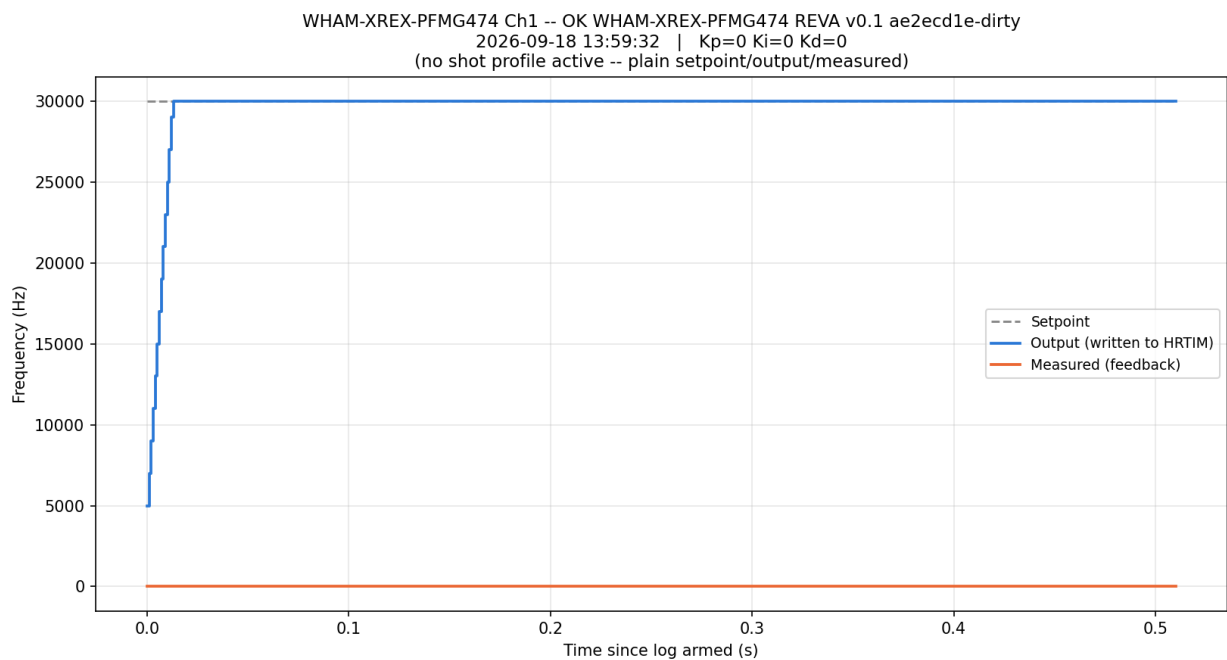
6 ENA_OUT/CONTACT_OUT dropped mid-shot

Expected: STATE? reports ENABLE_OUTPUT <ch>, survivor keeps running derated

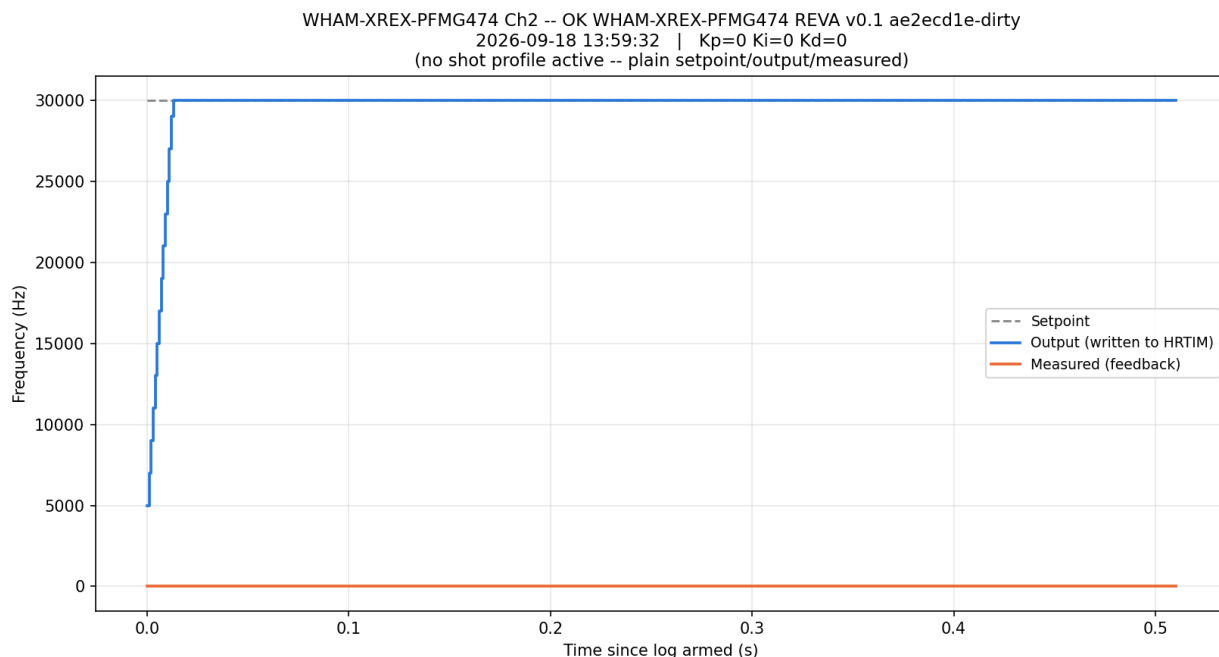
Verdict: PASS

Notes: STATE? -> 'OK FAULT ENABLE_OUTPUT 1'

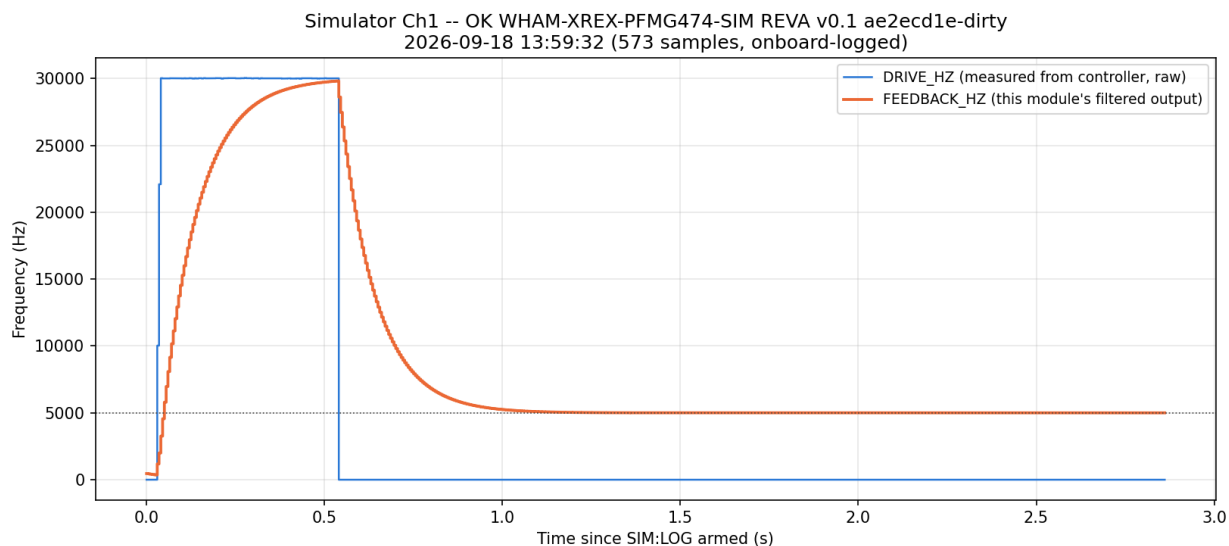
Raw data: shots/20260918_135929_ena_contact/ctrl_ch1.csv



Raw data: shots/20260918_135929_ena_contact/ctrl_ch2.csv



Raw data: shots/20260918_135929_ena_contact/sim_ch1.csv



7 External trigger regression (PF15 via DIAGnostic:GPOut12 loopback)

Expected: EXTERNAL:TRIGGER:INPUT? follows DIAGnostic:GPOut12 level

Verdict: **BLOCKED**

Notes: DIAGnostic:GPOut12 itself toggled correctly (low='OK 0' high='OK 1') but EXTERNAL:TRIGGER:INPUT? never followed (low='OK 0' high='OK 0') – the PD1->PF15 bench loopback wire appears disconnected, not a firmware issue.